

Supplementary Material for “Training Deep Neural Networks to Meet Hard Prediction Quotas”

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A Mechanism: Bounded vs. Unbounded Dual Behavior

Proof of Proposition 1. Write $u = E/K \geq 0$. In Eq. (1) the first term $E/(E + K) = u/(1 + u)$ increases from 0 to 1 and the second $\rho u^2/(1 + u^2)$ increases from 0 to ρ , so $0 \leq \ell < 1 + \rho$. Differentiating through $E = \max(0, S - K)$ gives Eq. (2) for $S > K$. There the first summand $K/(E + K)^2 = \frac{1}{K}(1 + u)^{-2}$ is at most $1/K$ (at $u=0$) and decreasing; the second, $\frac{2\rho}{K} u/(1 + u^2)^2$, is maximized at $u = 1/\sqrt{3}$ with value $\frac{2\rho}{K} \cdot \frac{9}{16\sqrt{3}} = \frac{9\rho}{8\sqrt{3}K} < \rho/K$ and tends to 0 as $u \rightarrow \infty$. Their sum therefore lies in $[0, (1 + \rho)/K]$ and vanishes as $E \rightarrow \infty$. Scaling ℓ and $\partial\ell/\partial S$ by a frozen multiplier λ gives the bounds $\lambda(1 + \rho)$ and $\lambda(1 + \rho)/K$, neither depending on E . $\square$

Why does a hard count cap stress the dual-ascent base-lines? On the paper grid the interaction is hidden: after the 50-epoch warmup the classifier is already saturated, and a saturation gate shared by every method (train accuracy ≥ 0.995 on two consecutive checks) stops cross-entropy updates within the first two constraint epochs, so the constraint phase is pure count optimization for all methods alike. We therefore probe with a one-epoch warmup, where the classifier keeps learning throughout the constraint phase and the two objectives visibly compete (Figure S1).

The probe shows three things. First, while the classifier is learning, the count cannot be forced down: each epoch’s cross-entropy updates re-inflate the constrained class, and the excess oscillates around its starting level for ~ 60 epochs under both methods (Fig. S1c). Dual ascent responds exactly as designed: the violation persists, so the multiplier grows every epoch, from 1 to 53 (seeds 2–3: 58, 60). In control terms this is integrator windup: the controller keeps integrating an error it has no ability to act on. Second, the escalation buys nothing. Every method takes a single gradient-clipped constraint step per epoch (max-norm 1.0, shared across methods), so extra multiplier cannot buy extra descent: the two cross-entropy traces are numerically indistinguishable (Fig. S1b), and the count comes down no faster at $\lambda=53$ than at TraLO’s ratcheted $\lambda \leq 0.19$. The asymmetry is in what the clip rail is doing. At $\lambda=53$ an unbounded-penalty gradient would dominate the update by orders of magnitude were it not clipped, whereas TraLO’s bounded penalty at $\lambda \leq 0.19$ demands orders of magnitude less force in the first place. Third, the count falls

only once the classifier saturates and the gate stops CE updates (dotted lines in Fig. S1c), and then both methods reach the cap within a few epochs. The count outcome is identical; what differs is the pressure applied to the constrained class to get there.

The paper grid shows the same behavior in compressed form: the dual multipliers escalate while TraLO’s freezes (≤ 0.09 there), Fioretto’s native satisfaction is ~ 0.87 (0.84 on OctMNIST and TissueMNIST, 0.93 on DermMNIST), so its unbounded multiplier stalls short of the cap on $\sim 13\%$ of runs that never reach feasibility within the budget (main paper Sec. 5.4); runs that end violated restore their best-satisfied checkpoint and remain competitive.

Nor is the escalation a step-size artifact. A dedicated fairness sweep (Block C of the released corpus; 96 runs on the hard-regime cells) varies Fioretto’s dual step over 0.001–0.01 and Hounie’s η over 0.001–0.1: Fioretto’s constrained-class F1 moves by less than 0.001 across the $10\times$ step range, larger Hounie steps only hurt ($0.43 \rightarrow 0.35$), and no setting closes the gap to TraLO on the OctMNIST tight cells (the best variant still trails by 0.019 cc-F1).

We state this mechanism *qualitatively only*. An audit shows that the magnitude of the escalation does not predict where TraLO wins (the multiplier is in fact largest in some cells where TraLO loses). So the claim is “dual ascent needs an unbounded multiplier plus a clip rail to hold a hard count, while TraLO’s bounded, frozen multiplier holds it alone,” not any dose-response of the form “more escalation $\Rightarrow$ bigger win.”

B Complete Component Ablation

A within-cell ablation isolates which ingredients carry the hard-regime advantage: on the OctMNIST tight-cap cells where TraLO leads, we disable one component at a time and measure the paired change in constrained-class F1 (Table S1; $n=24$ per component). Two components are load-bearing. Removing the optimizer reset at satisfaction costs $+0.079$ cc-F1, the single largest effect, helping on all 24 paired runs ($p = 3e-7$), because it breaks the post-satisfaction descent momentum that would otherwise erode the constrained class. Removing the undershoot hinge (the TraLO-bounded control, already in the frozen grid) costs $+0.036$ cc-F1 ($p = 3e-6$; 0.019–0.070 across cells). The other components are cc-F1-neutral on these already-satisfied cells:

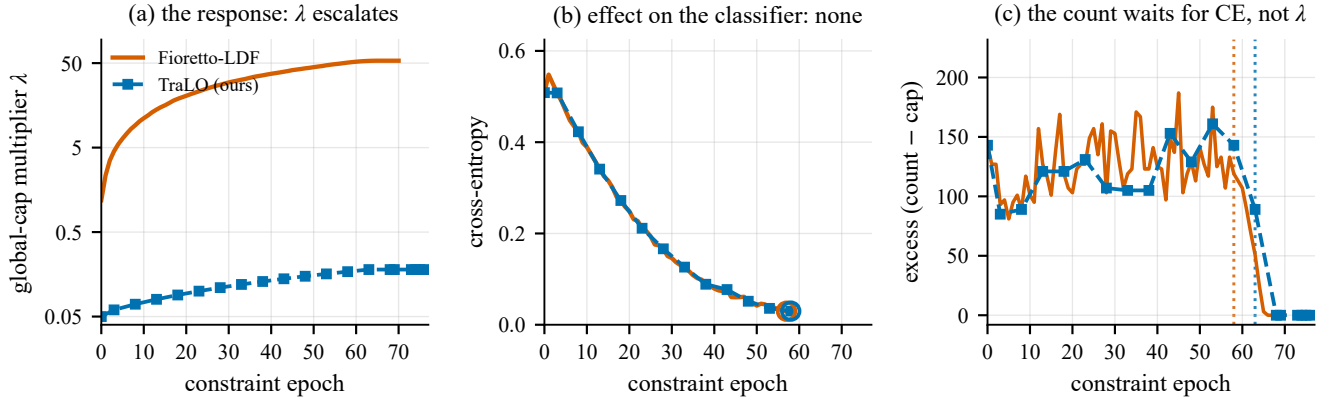


Figure S1: **Dual ascent winds up a multiplier the constraint cannot use: a one-epoch-warmup probe** (DermMNIST $L50/G50$, RegNetY-400MF, seed 1). While the classifier is still learning, the count cannot fall: each epoch’s CE updates re-inflate the constrained class, so the excess just oscillates (c, first ~ 60 epochs). Dual ascent reads that standing violation as “not enough pressure” and integrates it into $\lambda=53$ (a, log scale); the extra pressure changes nothing, since the constraint step is norm-clipped, so the CE traces stay numerically indistinguishable (b) and the count sits no lower at $\lambda=53$ than at TraLO’s ratcheted ≤ 0.19 . What frees the count is CE finishing (train-accuracy gate: circles in b, dotted lines in c); it then falls to the cap for both methods. Same outcome; the difference is the state each method arrives in: a wound-up multiplier held back by a clip rail, vs. a bounded nudge that never needed one. That state difference is what surfaces as stalls and missed feasibility on the main grid (main paper Fig. 4; Sec. A here). Each method keeps one multiplier per constraint; the local caps are never violated on this cell, so the local multipliers stay at zero and we plot the global-cap one. TraLO is logged every 5 epochs, hence its smoother-looking trace.

the ρ schedule and the lambda freeze move it by under 0.001 (n.s.). A dedicated KL sweep on TissueMNIST (the $L30$ tight cap, MobileNetV3, $\alpha_{KL} \in \{0, 0.1, 0.3, 1.0\}$) confirms the anchor is off for good reason: $\alpha_{KL}=0$ (the paper recipe) gives the best constrained-class *and* macro-F1, while every nonzero anchor lowers cc-F1 (by 0.01–0.03) and only adds prediction edits. The KL term was a useful drift damper in an early version of the method, before the optimizer reset and undershoot hinge; once those two control post-satisfaction drift directly, the anchor is redundant. The advantage rests on the reset and the hinge. The reset, ρ , and freeze legs are a dedicated ablation under the identical frozen recipe; the hinge leg is already in the main grid.

C Native Satisfaction: a Deployment Property, not a Headline

We separate *native satisfaction* (meeting the count with no post-hoc edit) from the quality results in the main paper because, on its own, it does not discriminate methods: every deployable system meets a hard count one way or another, and post-hoc clipping reaches satisfaction 1.00 by construction, simply by editing predictions. We report it only to document the mechanism behind TraLO’s quality advantage over clipping (main paper Sec. 5.3): TraLO reaches the count during training and so never invokes the editing step that costs clipping its accuracy. Table S2 gives the deployment costs at the tight $L30$ cap on MobileNetV3: macro-F1, accuracy, and the number of predictions each method must overwrite (Flips). Figure S2 shows native satisfaction per method over the full grid. TraLO and Hounie-RCL meet the count in every run;

Table S1: **Complete component ablation on the OctMNIST hard-binding cells.** Effect on constrained-class F1 of removing each TraLO component in turn ($n=24$ paired runs each: 2 tight caps $\times$ 3 backbones $\times$ 4 seeds). Δ cc-F1 = full TraLO – (component removed); winrate = fraction of the 24 paired runs where removing the component hurts; p is a two-sided paired test. Two components are load-bearing: the optimizer reset and the undershoot hinge, both helping on all 24 runs; the ρ schedule and lambda freeze are cc-F1-neutral on these already-satisfied cells. Two essential ingredients plus two neutral ones, *not* four co-equal components.

Component removed	Δ cc-F1	Winrate	p (paired)
<i>Load-bearing</i>			
Optimizer reset	+0.079	1.00	3e−7
Undershoot hinge	+0.036	1.00	3e−6
<i>Neutral</i>			
ρ schedule	+0.001	0.42	0.74
Lambda freeze	+0.000	0.38	0.65

Fioretto-LDF and the hinge-free TraLO-bounded do most of the time; the post-hoc clippers meet it natively only at loose caps. A caveat on *flips*: at near-trivial loose caps (e.g. OctMNIST $L80/L90$) TraLO can record more flips than a clipper that barely has to act, so the robust statement is “satisfies natively while preserving macro-F1,” never a raw flip count.

The macro-F1 advantage over clipping is not specific to MobileNetV3 or to the single $L30$ cap of Table S2. Table S3 averages the paired gap over the symmetric caps for

Table S2: **Model quality at the tight L30 cap** (MobileNetV3; each value is the mean over seeds 1–4 *at this single cap level*). All methods meet the count; the post-hoc clippers do so by *editing predictions*, and Flips is the number of test predictions overwritten to enforce the cap. This editing erodes **macro-F1**, the imbalance-aware quality metric, on every dataset. Raw *accuracy* is nearly blind to it: the constrained class is rare, so clipping it barely moves the aggregate and the clippers can tie or even beat the trained methods on accuracy while losing macro-F1. This is why F1, not accuracy, is the right quality metric here. [†] = post-hoc clipping. **Bold** = best, underline = 2nd-best per column within each dataset block.

Method	Tissue			Derm			OctMNIST		
	macro-F1 $\uparrow$	Acc. $\uparrow$	Flips $\downarrow$	macro-F1 $\uparrow$	Acc. $\uparrow$	Flips $\downarrow$	macro-F1 $\uparrow$	Acc. $\uparrow$	Flips $\downarrow$
Heur. [†]	0.347	0.478	99.5	0.722	0.846	136.5	0.666	0.695	109.5
Shifman [†]	0.347	0.478	99.5	0.721	0.846	136.5	0.665	0.695	109.5
Fioretto	0.405	0.536	7.8	0.723	0.842	10.2	<u>0.701</u>	0.738	<u>0.2</u>
Hounie	<u>0.406</u>	<u>0.537</u>	11.2	0.728	0.843	19.2	0.678	0.723	0.0
TraLO-b	0.405	0.536	10.0	<u>0.727</u>	<u>0.844</u>	<u>9.8</u>	0.700	0.738	0.0
TraLO	0.408	0.539	7.0	<u>0.726</u>	<u>0.844</u>	1.0	0.702	<u>0.736</u>	0.0

Table S3: **TraLO’s macro-F1 advantage over post-hoc clipping is backbone- and dataset-general.** Mean paired macro-F1 gap of TraLO over the best post-hoc clipper (per-seed best of Heuristic/Shifman-LP), averaged over the symmetric caps for each dataset–backbone cell; *win/tie/loss* counts the caps. The mean gain is positive in every cell, largest on TissueMNIST; it also holds under asymmetric caps (App. D).

Backbone	Tissue	Derm	OctMNIST
MobileNetV3	+0.058 9/0/0	+0.009 6/2/1	+0.044 9/0/0
RegNetY-400MF	+0.082 9/0/0	+0.005 6/2/1	+0.011 6/3/0
ViT-B/16	+0.020 9/0/0	+0.033 9/0/0	+0.007 5/4/0
MobileNetV2	+0.049 6/1/0	+0.004 3/3/0	+0.040 5/0/0

each dataset–backbone cell, adding MobileNetV2 to the three main-grid backbones: the mean gain is positive in every one of the twelve cells, largest on TissueMNIST, and it persists under asymmetric caps (App. D).

D Per-Cell Results Across the Full Grid

Tables S6–S9 give the fully granular breakdown behind the regime summary of the main paper (Sec. 5.2): one row per (dataset, backbone, cap), averaged over the four seeds *only*, with no pooling across backbones or caps. Each row pairs TraLO against the best constraint-trained baseline on constrained-class F1 (per-seed best of Fioretto-LDF/Hounie-RCL) and against the best post-hoc clipper on macro-F1, both paired by seed.

Two patterns read straight off the cells. The constrained-class win is *consistent* (all three backbones, every seed) only at the OctMNIST tight caps (L30/L40), where it is largest on ViT-B/16; a handful of isolated single-backbone cells elsewhere also clear the +0.005 bar, but no other regime wins across backbones, and none is a loss beyond seed noise. The macro-F1 margin over clipping is positive in 77 of the 81 symmetric cells; the four exceptions are the very tightest L10/L20 caps, where the cap nearly zeroes out the constrained class and the gap is within ± 0.02 . The asymmetric caps (Table S9) hold the *preliminary* constrained-class

Table S4: **The constrained-class picture is backbone-general.** Mean paired constrained-class-F1 gap of TraLO over the best constraint-trained baseline (per-seed best of Fioretto-LDF/Hounie-RCL), averaged over the symmetric caps for each dataset–backbone cell; *win/tie/loss* counts the caps (a win needs the mean gap and at least half the seeds to clear +0.005; loss is the mirror). Across all four backbones—the three in the main grid plus MobileNetV2—TissueMNIST and DermMNIST show no systematic constrained-class advantage (per-backbone means within ± 0.003 , mostly ties), while OctMNIST is a positive margin on every backbone. The honest tie and the OctMNIST win are both properties of the regime, not of one architecture.

Backbone	Tissue	Derm	OctMNIST
MobileNetV3	−0.002 0/6/3	+0.002 0/9/0	+0.004 2/6/1
RegNetY-400MF	−0.002 1/5/3	−0.003 0/7/2	+0.011 4/5/0
ViT-B/16	−0.001 1/7/1	−0.002 0/7/2	+0.018 5/4/0
MobileNetV2	+0.002 3/3/1	−0.002 1/3/2	+0.002 1/3/1

win, on DermMNIST where the global cap is tighter than the group cap. That per-cell table is MobileNetV3, the only backbone with the full asymmetric grid; a cross-backbone check on the four shared $G < L$ cap pairs (Table S5) shows the constrained-class win reproduces on RegNet (+0.9pp, 75% seed-winrate) and ties on the ViT (−0.0pp), while the macro-F1 advantage over clipping holds on all three backbones. We report the asymmetric result as preliminary because only MobileNetV3 has been run on the complete asymmetric grid, not because a single backbone shows it.

Table S4 summarizes the constrained-class picture across backbones, adding MobileNetV2 to the three main-grid backbones and averaging the paired gap over the symmetric caps: on TissueMNIST and DermMNIST no backbone shows a systematic advantage (means within ± 0.003), while OctMNIST is positive on all four. The honest tie and the OctMNIST win are each backbone-general, not artifacts of one architecture.

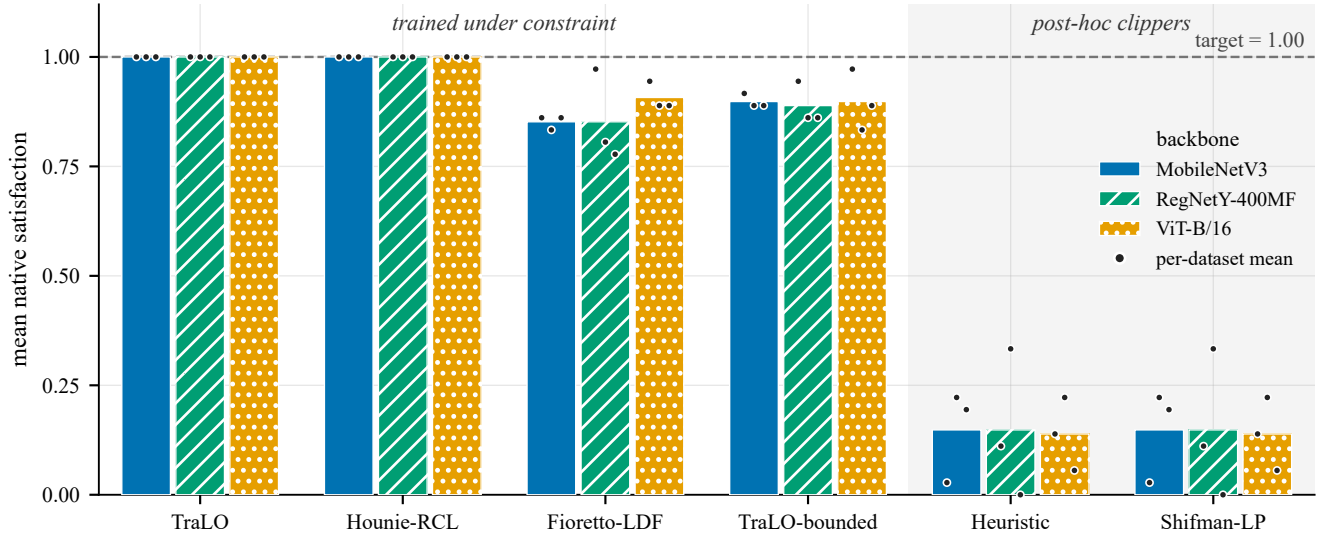


Figure S2: **Methods trained under the constraint satisfy the count natively; post-hoc clippers do not.** Mean native satisfaction per method and backbone over the paper_final grid ($N=108$ runs per method $\times$ backbone: 3 datasets $\times$ 9 caps $\times$ 4 seeds); the dots are the three per-dataset means. TraLO and Hounie-RCL meet the count in every run (satisfaction 1.00); Fioretto-LDF and TraLO-bounded reach it most of the time (~ 0.85 – 0.91); the post-hoc clippers satisfy natively only at loose caps (~ 0.14 – 0.15) and otherwise reach the cap solely by editing predictions, the source of their accuracy tax (Table S2).

Table S5: **The DermMNIST global-tighter ($G < L$) win reproduces across backbones.** Cross-backbone check on the four shared $G < L$ cap pairs ($L50/G20$, $L50/G30$, $L70/G50$, $L80/G70$; mean over seeds 1–4, $n=16$ paired seed-cells). $\Delta_{\text{cc-F1}}$ is TraLO minus the best constraint-trained baseline; $\Delta_{\text{macro-F1}}$ is TraLO minus the best post-hoc clipper; both paired by seed, with *win* the seed-cell fraction favoring TraLO. The constrained-class win holds on MobileNetV3 and RegNet and ties on the ViT; the macro-F1 advantage over clipping holds on all three.

Backbone	$\Delta_{\text{cc-F1 vs. trained}}$		$\Delta_{\text{macro-F1 vs. clipper}}$	
	Δ	win	Δ	win
MobileNetV3	+0.012	81%	+0.015	77%
RegNetY-400MF	+0.009	75%	+0.002	50%
ViT-B/16	−0.000	38%	+0.032	100%

Table S6: **TissueMNIST: per-cell constrained-class and overall F1.** One row per (backbone, cap), mean over seeds 1–4. cc-F1 columns compare TraLO to the best constraint-trained baseline (per-seed best of Fioretto-LDF/Hounie-RCL, paired); *win* is the seed-winrate. Δ_{mac} is TraLO minus the best post-hoc clipper on macro-F1. **Bold Δ** marks a paired win ($\Delta \geq 0.005$ on $\geq 3/4$ seeds).

Cap	cc-F1 vs. best trained				macro
	best	TraLO	Δ	win	Δ_{mac}
<i>MobileNetV3</i>					
L10	0.106	0.101	-0.005	1/4	+0.028
L20	0.205	0.205	-0.000	1/4	+0.054
L30	0.279	0.275	-0.005	1/4	+0.061
L40	0.301	0.300	-0.001	2/4	+0.060
L50	0.352	0.354	+0.002	2/4	+0.060
L60	0.381	0.375	-0.006	0/4	+0.064
L70	0.397	0.399	+0.002	2/4	+0.064
L80	0.414	0.407	-0.006	1/4	+0.065
L90	0.409	0.409	+0.000	1/4	+0.065
<i>RegNetY-400MF</i>					
L10	0.109	0.112	+0.003	1/4	+0.061
L20	0.212	0.195	-0.017	0/4	+0.073
L30	0.291	0.273	-0.018	0/4	+0.080
L40	0.314	0.324	+0.010	2/4	+0.091
L50	0.362	0.366	+0.004	3/4	+0.085
L60	0.379	0.383	+0.004	2/4	+0.084
L70	0.414	0.409	-0.005	1/4	+0.091
L80	0.429	0.432	+0.003	2/4	+0.089
L90	0.435	0.431	-0.005	1/4	+0.087
<i>ViT-B/16</i>					
L10	0.133	0.122	-0.011	0/4	+0.011
L20	0.239	0.239	+0.000	0/4	+0.016
L30	0.327	0.327	+0.000	0/4	+0.018
L40	0.360	0.373	+0.013	4/4	+0.020
L50	0.422	0.418	-0.004	0/4	+0.024
L60	0.445	0.447	+0.002	1/4	+0.023
L70	0.466	0.462	-0.003	1/4	+0.021
L80	0.487	0.482	-0.005	2/4	+0.025
L90	0.495	0.497	+0.002	2/4	+0.025

Table S7: **DermMNIST: per-cell constrained-class and overall F1.** One row per (backbone, cap), mean over seeds 1–4. cc-F1 columns compare TraLO to the best constraint-trained baseline (per-seed best of Fioretto-LDF/Hounie-RCL, paired); *win* is the seed-winrate. Δ_{mac} is TraLO minus the best post-hoc clipper on macro-F1. **Bold Δ** marks a paired win ($\Delta \geq 0.005$ on $\geq 3/4$ seeds).

Cap	cc-F1 vs. best trained				macro
	best	TraLO	Δ	win	Δ_{mac}
<i>MobileNetV3</i>					
L10	0.169	0.169	-0.000	1/4	-0.018
L20	0.310	0.312	+0.002	1/4	-0.001
L30	0.417	0.419	+0.002	1/4	+0.004
L40	0.506	0.510	+0.003	2/4	+0.012
L50	0.563	0.564	+0.002	1/4	+0.014
L60	0.618	0.618	-0.000	2/4	+0.013
L70	0.645	0.648	+0.003	3/4	+0.017
L80	0.655	0.660	+0.005	2/4	+0.022
L90	0.663	0.668	+0.005	3/4	+0.019
<i>RegNetY-400MF</i>					
L10	0.178	0.178	+0.000	0/4	-0.012
L20	0.317	0.317	+0.000	1/4	+0.004
L30	0.421	0.424	+0.003	3/4	+0.005
L40	0.502	0.500	-0.002	0/4	+0.006
L50	0.545	0.543	-0.002	1/4	+0.004
L60	0.598	0.588	-0.010	0/4	+0.010
L70	0.623	0.612	-0.011	1/4	+0.008
L80	0.640	0.637	-0.003	1/4	+0.011
L90	0.649	0.647	-0.002	1/4	+0.012
<i>ViT-B/16</i>					
L10	0.169	0.165	-0.004	0/4	+0.021
L20	0.300	0.293	-0.007	0/4	+0.029
L30	0.388	0.386	-0.002	0/4	+0.030
L40	0.460	0.460	+0.000	1/4	+0.034
L50	0.510	0.507	-0.003	0/4	+0.036
L60	0.546	0.546	+0.000	1/4	+0.035
L70	0.575	0.575	+0.000	2/4	+0.028
L80	0.610	0.604	-0.006	0/4	+0.037
L90	0.606	0.612	+0.006	2/4	+0.043

Table S8: **OctMNIST: per-cell constrained-class and overall F1**. One row per (backbone, cap), mean over seeds 1–4. cc-F1 columns compare TraLO to the best constraint-trained baseline (per-seed best of Fioretto-LDF/Houinier-RCL, paired); *win* is the seed-winrate. Δ mac is TraLO minus the best post-hoc clipper on macro-F1. **Bold** Δ marks a paired win ($\Delta \geq 0.005$ on $\geq 3/4$ seeds).

Cap	cc-F1 vs. best trained				macro
	best	TraLO	Δ	win	Δ mac
<i>MobileNetV3</i>					
L10	0.176	0.178	+0.002	1/4	+0.037
L20	0.312	0.312	+0.000	1/4	+0.044
L30	0.391	0.407	+0.016	4/4	+0.036
L40	0.487	0.509	+0.022	4/4	+0.053
L50	0.581	0.583	+0.001	2/4	+0.050
L60	0.635	0.639	+0.004	2/4	+0.049
L70	0.658	0.650	-0.008	1/4	+0.041
L80	0.676	0.677	+0.001	1/4	+0.042
L90	0.689	0.692	+0.002	1/4	+0.045
<i>RegNetY-400MF</i>					
L10	0.171	0.173	+0.002	1/4	-0.001
L20	0.308	0.308	+0.000	1/4	+0.007
L30	0.372	0.407	+0.035	4/4	+0.006
L40	0.452	0.473	+0.021	4/4	+0.006
L50	0.545	0.544	-0.001	1/4	+0.011
L60	0.583	0.603	+0.020	3/4	+0.016
L70	0.626	0.645	+0.019	4/4	+0.018
L80	0.647	0.652	+0.006	3/4	+0.014
L90	0.672	0.671	-0.001	1/4	+0.018
<i>ViT-B/16</i>					
L10	0.176	0.175	-0.002	0/4	+0.000
L20	0.313	0.318	+0.005	2/4	+0.003
L30	0.341	0.422	+0.081	4/4	+0.004
L40	0.462	0.512	+0.051	4/4	+0.005
L50	0.592	0.597	+0.005	4/4	+0.010
L60	0.638	0.643	+0.005	3/4	+0.007
L70	0.671	0.672	+0.001	1/4	+0.010
L80	0.697	0.701	+0.004	2/4	+0.013
L90	0.701	0.714	+0.013	4/4	+0.011

Table S9: **Asymmetric caps (MobileNetV3): per-cell F1**. Global cap L and group cap G set independently; one row per cap pair, mean over seeds 1–4. Columns as in Table S7. The global-tighter ($G < L$) block is where the single preliminary cc-F1 win sits; Δ mac stays positive throughout.

Cap (L/G)	cc-F1 vs. best trained				macro
	best	TraLO	Δ	win	Δ mac
<i>DermMNIST, $G < L$ (global tighter)</i>					
30/20	0.296	0.313	+0.017	3/4	-0.001
50/20	0.295	0.314	+0.019	3/4	+0.029
50/30	0.411	0.417	+0.006	2/4	+0.002
70/20	0.296	0.306	+0.009	2/4	+0.006
70/30	0.413	0.417	+0.005	2/4	+0.006
70/50	0.554	0.565	+0.011	4/4	+0.017
80/20	0.295	0.287	-0.007	1/4	+0.003
80/30	0.411	0.410	-0.001	1/4	+0.033
80/50	0.553	0.563	+0.010	4/4	+0.015
80/70	0.639	0.650	+0.010	4/4	+0.024
<i>DermMNIST, $G > L$ (local tighter)</i>					
20/30	0.312	0.312	+0.000	1/4	+0.003
20/50	0.312	0.310	-0.002	1/4	+0.028
20/70	0.312	0.310	-0.002	0/4	+0.001
20/80	0.312	0.310	-0.002	0/4	+0.003
30/50	0.416	0.417	+0.002	2/4	+0.012
30/70	0.417	0.416	-0.002	1/4	+0.010
30/80	0.417	0.417	+0.000	1/4	+0.018
50/70	0.567	0.561	-0.006	0/4	+0.016
50/80	0.564	0.558	-0.006	0/4	+0.018
70/80	0.645	0.646	+0.001	3/4	+0.025
<i>TissueMNIST, $G < L$ (global tighter)</i>					
30/20	0.171	0.172	+0.001	1/4	+0.015
50/20	0.177	0.166	-0.011	0/4	+0.023
50/30	0.235	0.249	+0.014	3/4	+0.015
70/20	0.179	0.166	-0.014	0/4	+0.014
70/30	0.236	0.216	-0.021	0/4	+0.011
70/50	0.297	0.297	+0.000	2/4	+0.016
80/20	0.179	0.164	-0.015	0/4	+0.013
80/30	0.238	0.245	+0.007	3/4	+0.026
80/50	0.298	0.292	-0.006	2/4	+0.012
80/70	0.324	0.329	+0.005	2/4	+0.016
<i>TissueMNIST, $G > L$ (local tighter)</i>					
20/30	0.188	0.185	-0.002	0/4	+0.011
20/50	0.190	0.178	-0.012	0/4	+0.024
20/70	0.188	0.185	-0.002	1/4	+0.012
20/80	0.188	0.183	-0.005	1/4	+0.008
30/50	0.243	0.243	+0.000	2/4	+0.018
30/70	0.243	0.241	-0.002	1/4	+0.016
30/80	0.243	0.243	+0.000	1/4	+0.026
50/70	0.303	0.317	+0.014	3/4	+0.019
50/80	0.298	0.313	+0.016	3/4	+0.016
70/80	0.340	0.345	+0.005	2/4	+0.018